

Practical Assignment

Learning Goals

1. Gain a deep understanding of an advanced topic in ML by means of a literature review.
2. Understand the methodology for development of innovative solutions for advanced ML topics.

Description

In this project the students will develop a research project on an advanced ML topic of their choice. The goal is understand how reliable, innovative solutions should be developed for advanced ML problems. Given a paper of their choice, the students will investigate a limitation of the corresponding work. This involves a short but systematic literature review on the topic and an empirical study concerning the contribution in the selected paper. Based on this, students will propose an alternative solution and then empirically compare it to the original approach, to assess if the limitations identified have been addressed. The results of the project will be discussed with the course lecturers.

Workplan

Phase 1 – paper selection

- Select a paper on an advanced ML topic, in collaboration with the course lecturers.
 - The paper should include code, to minimize the implementation effort.
 - Students are encouraged to select papers from topics addressed in the course, but this is not mandatory.
 - Students are encouraged to select recent papers from important conferences and journals in ML (e.g., NeurIPS, KDD, ICML, ICLR, ECMLPKDD, Discovery Science, Intelligent Data Analysis, Machine Learning journal, Journal of Machine Learning Research, Data Mining and Knowledge Discovery journal, International Journal of Data Science and Analytics).

Phase 2 – analysis of selected contribution

- Analyze the contribution of the selected paper to identify an interesting and relevant limitation. The analysis method should be discussed with the course lecturers but it usually involves:

- Empirical evaluation of the contribution of the paper on a set of benchmark problems.
- Literature review, focused on the limitation of interest.
- Given the limited time for the project, this analysis may have to be simplified.
- Depending on the project, other methods may be more suitable.

Phase 3 – development of an alternative approach

- Propose a variant that is expected to address the limitation identified.
- Implement the proposed variant.
- Empirically evaluate if the proposed variant adequately addresses the limitation identified, using the same set of benchmark problems used to evaluate the original contribution.

Evaluation moments

- **Checkpoint 1:** Discussion of paper selected for project. This consists of a 3min presentation followed by a 5min discussion with the course lecturers.
- **Checkpoint 2:** Discussion of problem addressed and planned solution. This consists of a 5min presentation followed by a 10min discussion with the course lecturers.
- **Project submission:** Your assignment should be submitted in Moodle with a compressed file containing the following items:
 - the source of a ready-to-execute notebook with all the code necessary to run to obtain the presented results, including any complementary files needed to execute your notebook (e.g. data files, data objects);
 - slides for presentation (PDF format) focusing on the main issues of the assignment; any additional information that cannot be presented in that time slot can be included as annexes to the presentation; the total number of slides in the final submission, including annexes, should not exceed 40.
- **Final project discussion:** This consists of an 8min presentation followed by a 12min discussion with the course lecturers.

All checkpoints are mandatory. Any missing evaluation moment will be graded with 0. The specific dates of these moments are indicated in the course schedule.

Groups

The practical assignment is mandatory and should be performed by **groups of three students**. Please select your group in Moodle **until the deadline indicated in the course schedule**.

Grading details

- **Checkpoint 1 - (10%)**
 - **Paper selection**
 - Relevance of the ML topic
 - Quality of the paper
- **Checkpoint 2 - (25%)**
 - **Comprehension of the selected topic**
 - Description of the problem
 - Description of the contribution
 - Discussion of the limitation
 - Literature review
 - **Empirical study**
 - experimental setup
 - analysis of results
 - **Proposal to address the limitation**
 - theoretical and empirical motivation
 - description
 - originality
- **Final presentation & submission - (50%)**
 - **Comprehension of the selected topic**
 - Description of the problem
 - Description of the contribution
 - Discussion of the limitation
 - Literature review
 - **Proposal to address the limitation**
 - theoretical and empirical motivation
 - description
 - originality
 - **Empirical study (20%)**
 - experimental setup
 - analysis of results
 - **Notebook**
 - organization and python implementation
 - **Presentation**
 - slides
 - delivery
- **Individual grade (15%)**
 - An individual grade will be given to each student based on the discussion.